# AE 5335 Midterm

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Numerical

| Case | T (K) | Tt (K) | P | Pt | u (m/s) | M | F (N) | q_conv (J/kg) |
|---|---|---|---|---|---|---|---|---|
| 1 | 313.258 | 316.224 | 135988.135 | 140548.282 | 77.193 | 0.218 | 19166.918 | |
| 2 | 504.026 | 759.104 | 29938.983 | 125518.008 | 715.857 | 1.591 | -0.004 | |
| 3 | 364.483 | 560.000 | 24728.928 | 111171.308 | 626.733 | 1.638 | -1841.481 | |
| 4 | 1746.892 | 1749.718 | 197169.092 | 198287.693 | 75.347 | 0.090 | -42.521 | 1556655.099 |
| 5a | 817.299 | 823.260 | 3898002.021 | 3998422.032 | 109.435 | 0.191 | 28590.761 | |
| 5b | 806.084 | 823.260 | 2265082.490 | 2438560.427 | 185.762 | 0.326 | 13649.934 | |
| 6a | 896.638 | 950.138 | 49539.601 | 60680.126 | 327.845 | 0.546 | -844.129 | |
| 6b | 877.257 | 950.136 | 41530.073 | 54912.156 | 382.640 | 0.644 | -1117.137 | |
| 7 | 560.789 | 1876.166 | 5645.890 | 386708.157 | 1625.605 | 3.425 | 4851.971 | -200253.544 |

Analytical

| Case | T (K) | Tt (K) | P | Pt | u (m/s) | M | F (N) |
|---|---|---|---|---|---|---|---|
| 1 | 313.258 | 316.224 | 135987.767 | 140548.043 | 77.194 | 0.218 | 19166.888 |
| 2 | 504.027 | 759.104 | 29939.094 | 125518.046 | 715.857 | 1.591 | 0.000 |
| 3 | 364.483 | 560.000 | 24728.997 | 111171.561 | 626.732 | 1.638 | -1841.469 |
| 4 | | | | | | | |
| 5a | 817.297 | 823.260 | 3897947.020 | 3998389.201 | 109.448 | 0.191 | 28590.594 |
| 5b | | | | | | | |
| 6a | 896.629 | 950.136 | 49539.010 | 60680.881 | 327.863 | 0.546 | -844.036 |
| 6b | | | | | | | |
| 7 | | | | | | | |

Percent Difference

| Case | T (K) | Tt (K) | P | Pt | u (m/s) | M | F (N) |
|---|---|---|---|---|---|---|---|
| 1 | 9.72E-05 | 6.85E-05 | 2.70E-04 | 1.70E-04 | 1.48E-03 | 1.53E-03 | 1.57E-04 |
| 2 | 8.48E-05 | 1.22E-05 | 3.70E-04 | 3.04E-05 | 1.02E-04 | 1.44E-04 | NaN |
| 3 | 2.97E-05 | 4.45E-05 | 2.79E-04 | 2.28E-04 | 3.60E-05 | 2.11E-05 | 6.68E-04 |
| 4 | | | | | | | |
| 5a | 1.98E-04 | 2.99E-05 | 1.41E-03 | 8.21E-04 | 1.15E-02 | 1.16E-02 | 5.85E-04 |
| 5b | | | | | | | |
| 6a | 9.65E-04 | 2.68E-04 | 1.19E-03 | 1.24E-03 | 5.70E-03 | 6.18E-03 | 1.10E-02 |
| 6b | | | | | | | |
| 7 | | | | | | | |

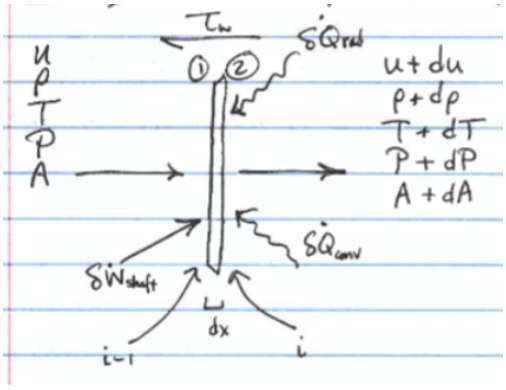



**Fig. 1 Quasi 1D flow**

# I. Numerical Methodology

### A. quasi 1D flow solver

For this assignment a code was developed to integrate the quasi-1D flow equations using the callorically perfect gas assumption. The equations of state are:

$$\frac{d\rho}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0 \tag{1}$$

$$\frac{dP}{\rho} + udu = \frac{-\tau_w c dx}{\rho A} + \eta \delta w_{shaft} \tag{2}$$

$$C_p dT + udu = \delta q + \delta w_{shaft} \tag{3}$$

$$\frac{dP}{P} = \frac{d\rho}{\rho} + \frac{dT}{T} \tag{4}$$

In this current form we are left with a system of 4 equations with 4 unknowns which can be solved for using a root solving algorithm. However, since root solvers themselves can be computationally expensive and introduce numerical additional numerical errors an additional step was made. The system of equations shown above was reduced algebraically to a single equation with one unknown. The variable that was solved for was $du$ due to its presence in the continuity, momentum, and energy equations.

$$du = -\frac{\frac{\rho}{P}\left(\eta dw - \frac{\tau_w c dx}{\rho A}\right) - \frac{dq+dw}{CpT} + \frac{dA}{A}}{\frac{1}{u} - \frac{u}{RT} + \frac{u}{CpT}} \tag{5}$$

The code itself will take the following inputs: an array with the velocity, temperature, and pressure.; A function defining the area of the stream tube with respect to x; length of the stream tube; name of the working fluid; number of iterations; work per mass through the stream tube; heat per mass through the stream tube; skin friction coefficient; work efficiency; and wall temperature.

After entering these inputs and pre-generating arrays, equation 5 was used to solve for $du$. This was then entered into equations 2 and 3 to find $dP$ and $dT$. This was then plugged into equation 4 to solve for $d\rho$. $dw$ is defined as the total work divided by the number of axial steps and $dq$ for circular ducts is defined as:

$$dq = dq_{conv} + dq_{rad} = 2C_p C_f \left(T_w - T_t\right) \frac{dx}{D} + \frac{q}{N} \tag{6}$$

Using this methodology, calculating the fluid dynamics through a stream tube using the quasi 1D approach with 1 million steps takes 4.9 seconds. As shown in the results, it is able to do this while only having small numerical differences with the analytical solution.

### B. Additional Equations

$$F = \dot{m}\,(u_e - u_i) + P_e A_e - P_i A_i \tag{7}$$

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2 \tag{8}$$

$$\frac{P_t}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{9}$$

$$C_p T_t = C_p T + \frac{u^2}{2} \tag{10}$$

## II. Problem Descriptions and Analytical Methodology

**Case 1: Isentropic Diverging Subsonic Nozzle**

This problem uses Mach area relations. With these two equations and the use of a root finder the Mach number at the exit can be found.

$$G(M) = \frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma + 1}\left(1 + \frac{\gamma - 1}{2} M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{11}$$

$$\frac{A_e}{A_i} = \frac{G(M_e)}{G(M_i)} \rightarrow G(M_e) \rightarrow M_e \tag{12}$$

Since the flow is isentropic the total temperature and pressure are constant. Using the exit Mach number and inlet total temperature and pressure, the conditions at the exit can be found.

**Case 2: Rayleigh Flow Tube**

Using equations for constant area heat addition, first the total temperature at the exit can be found. When heat is added to a constant area tube one should expect to see total temperature increase and total pressure drop. The flow is supersonic at the inlet so through the tube the Mach number should heat towards 1. Using the ratio of the total temperatures from the inlet and exit as well as a numerical solver (this can also be solved analytically) the exit Mach number can be found.

$$f(M) = \left[\frac{1 + \frac{\gamma-1}{2} M^2}{\left(1 + \gamma M^2\right)^2}\right] \tag{13}$$

$$\frac{T_{te}}{T_{ti}} = 1 + \frac{q}{C_p T_{t1}} = \frac{f(M_e)}{f(M_i)} \rightarrow f(M_e) \tag{14}$$

The exit pressure can then be found by using the following equation.

$$P_e = P_i \frac{1 + \gamma M_i^2}{1 + \gamma M_e^2} \tag{15}$$

This is enough information to find the rest of the conditions for the exit.

**Case 3: Fanno Flow Tube**

This problem involves a constant area adiabatic tube with no heat addition and has viscous effects. Because the flow is adiabatic one should expect to see a constant total temperature, though since there are irreversibility introduced there will be a drop in total pressure. Since the flow is supersonic the Mach number should also move downward towards 1. To solve for the exit Mach number the following two equations can be used.

$$f_v(M) = \frac{\gamma + 1}{2} \ln \frac{1 + \frac{\gamma-1}{2} M^2}{M^2} - \frac{1}{M^2} \tag{16}$$

$$f_v(M_e) = \gamma C_f \frac{c}{A} L + f_v(M_i) \tag{17}$$

By setting the second equation equal to the first, a root finding algorithm can be used to find the exit Mach number. Now that the exit Mach number is known and that the flow is adiabatic, the following can be said.

$$\frac{P_e}{P_i} = \frac{M_i}{M_e} \left[ \frac{1 + \frac{\gamma-1}{2} M_i^2}{1 + \frac{\gamma-1}{2} M_e^2} \right]^{1/2} \tag{18}$$

Knowing the exit Mach number, total temperature, and static pressure the rest of the exit conditions can be found.

**Case 4: Convective Heat Transfer**

This problem has a hot wall transferring heat through convection into a fluid with a lower static temperature. This convective heat transfer starts with the interaction between the boundary layer and the wall, so this problem is a viscous problem.
No analytical solution is given for this problem

**Case 5 and 6: Variable Area Work Interaction**

These problems involve positive and negative work interactions within a tube with changing area. For the $\eta = 1$ case the following three equations can be used to calculate the exit conditions. The first is a form of the energy equation rewritten to solve for the exit total temperature. The second equation relates the ratio of total pressures to total temperatures using the ratio of specific heats.

$$T_{te} = \frac{w}{C_p} + T_{ti} \tag{19}$$

$$P_{te} = P_{ti} \left( \frac{T_{te}}{T_{ti}} \right)^{\frac{\gamma}{\gamma-1}} \tag{20}$$

Modifying the mass flow rate equation with the ideal gas equation and equation 8 and 9 gives the following.

$$\dot{m} = P_{te} M_e A_e \sqrt{\frac{\gamma}{R T_{Te}}} \left( 1 + \frac{\gamma - 1}{2} Me^2 \right)^{\frac{-(\gamma+1)}{2(\gamma+1)}} \tag{21}$$

It can be assumed that the mass flow rate is constant through the tube. This meaning the mass flow at the inlet is the same as the exit. Knowing this, the equation above can be set equal to 0 using the inlet mass flow and then the exit Mach number can be found. Knowing the exit total temperature, total pressure, and Mach number is enough to compute the rest of the exit conditions.

**Case 7: Variable Area Work Interaction with Convective Heat Transfer**

This problem is a combination of the previous problems. There is a work interaction with an associated efficiency, viscous cooling from the wall, and expansion of supersonic flow.
No analytical solution is given for this problem